# Step 2 — what the work is, and what to automate

*Generated by* `src/make_step2_report.py` *from* `out/segments.jsonl`*. Do not edit by hand — an earlier hand-maintained version silently went three pipeline revisions out of date.*

All figures are **relative**. The brief states the recordings were made in a test environment with compressed waiting times, so absolute durations are meaningless and only comparisons between processes are valid.

## Scale

**664 executions, 173 minutes of observed work, 15 sessions, 4 operators**, all on 2026-07-01.

### How many people

| source | count |
|---|---|
| `username_hash` | 4 |
| `machine_id` | 4 |
| operator names on the portal dashboard | 3 |

Where each of those names appears. The names themselves are withheld; each row is keyed by the system the name belongs to.

| name belongs to | machines it appears on | sessions it appears in | appearances in that system |
|---|---|---|---|
| fin | 4 of 4 | 15 of 15 | 100% |
| hr | 4 of 4 | 15 of 15 | 100% |
| ops | 4 of 4 | 13 of 15 | 100% |

The three names are **not** operators — each appears on every machine, and each belongs to one portal system. They are shared per-system logins. So: **4 people** working across three systems under shared accounts, which is a governance finding rather than a trivial one — automation would run with no per-user audit trail.

## The route names describe nothing

The portal is one SPA deployed three times, so its route names repeat. Each screen prints its own title, which names the work; the document most often open during it shows the kind of case handled there:

| segment label | dominant document | segments | purity |
|---|---|---|---|
| `ops__leave-applications` | keiyaku_kaijo_tetsuzuki | 52 | 65% |
| `hr__onboarding` | nyusha_checklist_shinsotsu_batch | 51 | 80% |
| `fin__payroll-items` | getsujitsu_teigaku_torihikisaki_ichiran | 50 | 94% |
| `fin__onboarding` | gyomu_itaku_kyuuyo_kitei | 30 | 97% |
| `fin__leave-applications` | settai_keihi_kitei | 27 | 85% |
| `fin__resident-tax` | shinkuitorihikisaki_touroku_tetsuzuki | 20 | 90% |
| `hr__payroll-items` | gyomu_itaku_keihi_kitei | 18 | 56% |
| `hr__social-insurance` | ikuji_kyuugyou_kitei | 17 | 59% |
| `ops__social-insurance` | kanrisya_kengen_shinsei_tetsuzuki | 7 | 100% |
| `hr__leave-applications` | keiyaku_kaijo_tetsuzuki | 2 | 50% |

Weighted document purity across labels: **80.4%**. The pipeline never reads document names, so this is independent evidence that the groupings track real business processes.

## Ranking

Two measurable quantities in tension: **mechanical load** (clipboard events and application switches per run — what automation removes) against **judgment load** (share of runs with a regulation document open — what it does not). High volume with high judgment is a poor first target, because the residue is what costs the time.

Priority is the hand transfers automation would remove, discounted for judgment: executions × transfers per run × (1 − judgment share). The table is in that order, computed before rounding; recomputed from the rounded figures shown, near-tied neighbours can change places.

| process | n | min | share | median | mech | judgment | people |
|---|---|---|---|---|---|---|---|
| expense_and_pay_change | 120 | 27.2 | 15.8% | 11 s | 2.6 | 15% | 4 |
| inventory_management | 63 | 15.5 | 9.0% | 11 s | 3.1 | 3% | 4 |
| purchase_orders | 73 | 15.6 | 9.0% | 11 s | 2.1 | 27% | 4 |
| attendance_and_leave | 58 | 13.2 | 7.7% | 10 s | 2.0 | 3% | 3 |
| invoice_approval | 77 | 23.4 | 13.6% | 16 s | 3.9 | 65% | 4 |
| it_requests | 42 | 8.5 | 4.9% | 11 s | 2.7 | 17% | 4 |
| benefits_applications | 35 | 7.7 | 4.5% | 13 s | 3.0 | 49% | 3 |
| onboarding_procedures | 58 | 19.2 | 11.1% | 19 s | 6.2 | 88% | 4 |
| contract_management | 59 | 19.1 | 11.1% | 19 s | 6.2 | 88% | 4 |
| payments | 37 | 11.0 | 6.4% | 17 s | 5.1 | 81% | 3 |
| manager_expense_approval | 34 | 9.5 | 5.5% | 16 s | 4.9 | 79% | 2 |
| budget_variance_analysis | 8 | 2.7 | 1.6% | 21 s | 3.5 | 0% | 1 |

### Robustness

A priority formula is easy to make say what you want, so the ranking was scored under 5 different weightings. Processes with n < 5 were excluded, since per-run rates on one or two observations are noise.

| process | top-3 appearances |
|---|---|
| **expense_and_pay_change** | **5 / 5** |
| invoice_approval | 2 / 5 |
| onboarding_procedures | 2 / 5 |
| purchase_orders | 2 / 5 |
| inventory_management | 2 / 5 |
| attendance_and_leave | 1 / 5 |
| contract_management | 1 / 5 |

Only **expense_and_pay_change** survives every weighting. Anything appearing once is an artefact of a particular formula, not a finding.

## The result that sets the scope

Aggregating by portal **screen** rather than by system. Judgment is the share of the screen's runs with a regulation document open:

| screen | executions | minutes | share of all work | systems | judgment |
|---|---|---|---|---|---|
| **payroll-items** | **260** | **66.1** | **38.3%** | **3** | 27% |
| leave-applications | 151 | 41.8 | 24.2% | 3 | 54% |
| onboarding | 95 | 30.1 | 17.5% | 2 | 85% |
| social-insurance | 85 | 18.9 | 11.0% | 3 | 28% |
| resident-tax | 73 | 15.6 | 9.0% | 1 | 27% |

**One screen pattern accounts for 38.3% of all observed work**, occurs in 3 systems, and carries the lowest judgment load of the five screens, though only just: 26.9%, against 27.4% for resident-tax and 28.2% for social-insurance. The two top-ranked processes are this one screen in different deployments.

That reframes Step 3: the choice is not *which process to automate* but *one bespoke process, or the shared pattern behind several* — which is exactly the scope question the brief says is itself part of the ROI decision.